TM TradeMind

Systems Proposal

An online marketplace for expert-led lessons, subscriptions, and private tutoring

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Team 5 · BCIS 4610.001

Business case · Feasibility · UML models · Data model · Interface prototypes · Usability

THE PROJECT

What TradeMind is

An online educational marketplace where users buy individual lessons, subscribe for unrestricted access to all content, and book private tutoring with verified experts. The platform is organized, thoroughly moderated, and accessible to learners and content creators alike.

Customers

- Browse lessons and review instructor profiles
- Pay securely online
- Manage subscriptions
- Access a library of purchased content

Subject-matter experts

- Upload and manage lesson materials
- Monitor earnings and analytics
- Schedule tutoring availability

TradeMind staff

- Approve and moderate lessons
- Oversee daily operations
- Manage platform financials

System scope. Lesson purchases, subscriptions, tutoring appointments, user accounts, payment processing, content moderation, and platform administration. The project covers business case, requirements and feasibility analysis, UML modeling, Unified Process scheduling, data modeling, interface design, and usability testing.

BUSINESS CASE

Business model and pricing

$19.99 per lesson

For learners who want one or two skills

$45 per month

Full access to all content

$359 per year

Annual subscription at a discount

Tutoring and rebates

- Private tutoring: $40 for non-subscribers, $25 for subscribers
- Subscribers who review unreleased demo lessons earn rebates of up to $10 per month

Expert compensation

- Upfront payment for approved content
- Ongoing percentage of sales and views, rewarding quality and self-promotion

Quality control

- Employees vet every lesson before it is posted
- Protects credibility and prevents misinformation

FEASIBILITY

Technical and operational

Technical: medium risk

- Web-based platform; the focus is a well-optimized website rather than a mobile app
- Built by contracted web developers on Amazon Web Services for video hosting, security, servers, and databases
- A contracted maintenance service covers later technical problems
- Existing payment processor and cloud-hosted database for accounts and payments
- Two to six months to reach the testing phase
- All required technology already exists; the risk is integrating security, video streaming, recurring payments, and large servers

Operational

- A moderated digital marketplace on a proven format, with stronger content oversight than competitors
- Content review team vets creators and approves lessons before publication
- Staff: Platform Operations Manager, IT support, customer service representatives, content reviewers
- Automated revenue sharing: upfront payment plus a percentage of each sale

FEASIBILITY

Schedule: Unified Process, 10–12 months

Inception	Elaboration	Construction	Transition
Months 1–2	Months 3–5	Months 6–9	Months 10–12

Inception

- Identify stakeholders
- Feasibility analyses
- Requirements gathering
- Preliminary plan and budget

Elaboration

- Systems requirements
- Use case, sequence, state chart models
- User interface design

Construction

- Lesson management
- Tutoring scheduling
- Subscription and payment services
- Database integration

Transition

- User testing and deployment
- Creator onboarding
- Finishing touches
- Post-implementation support

Delays could come from technical challenges or extended testing; structured project management keeps the timeline achievable.

ORGANIZATIONAL FEASIBILITY

Legal, political, and risk

Legal and contractual

- Copyright, data privacy, and digital consumer protection compliance
- Creator Terms of Service bar redistribution; DRM and screen-recording limits where possible
- 18+ age gate, or parental consent where applicable
- Copyright attorney retained; independent-contractor agreements cover pay, confidentiality, and IP

Political

- Stricter digital copyright enforcement could require more content monitoring
- Privacy rules could raise compliance costs
- Tax changes could shift revenue and pricing
- Contractor reclassification could raise payroll expense
- Mitigation: monitor legislation and keep policies adaptable

Risk analysis

- Misinformation if vetting is inconsistent
- Hacking, or unauthorized recording and redistribution of paid lessons
- Demand below projections, or low-quality creator content
- Variable contractor performance
- Mitigation: strong review, cybersecurity, clear contracts, ongoing monitoring

ECONOMIC FEASIBILITY

Five-year cost-benefit: NPV $1.54M

$3,558,314.86

Present value of benefits

$2,015,301.69

Present value of costs

$1,543,013.17

Net present value

Revenue streams

- $19.99 pay-per-lesson
- $45 monthly subscription
- $359 annual subscription
- Private tutoring: $40 non-subscribers, $25 subscribers

Assumptions

- 3% discount rate, kept conservative
- Revenue grows about 10% a year with industry demand and user growth
- Costs grow about 5% a year (hosting, staffing, marketing)
- Major costs: development, AWS hosting, payment fees, marketing, content reviewers, lesson-provider pay

Cost-benefit analysis

Cost Benefit Analysis

	Current Year (CY)	CY +1	CY +2	CY +3	CY +4	CY +5	
Costs							
Platform Development	$ 140,000.00	$ 20,000.00	$ 15,000.00	$ 15,000.00	$ 15,000.00	$ 15,000.00	
Legal & Licensing	$ 18,000.00	$ 5,000.00	$ 5,250.00	$ 5,513.00	$ 5,789.00	$ 6,078.00	
Initial Marketing Launch	$ 60,000.00	$ -	$ -	$ -	$ -	$ -	
Office & Equipment	$ 20,000.00	$ 5,000.00	$ 6,000.00	$ 5,500.00	$ 5,750.00	$ 6,500.00	
Salaries & Benefits	$ 60,000.00	$ 180,000.00	$ 189,000.00	$ 198,450.00	$ 208,373.00	$ 218,792.00	
Hosting & Tech	$ 25,000.00	$ 36,000.00	$ 37,800.00	$ 39,690.00	$ 41,675.00	$ 43,759.00	
Ongoing Marketing	$ 15,000.00	$ 60,000.00	$ 63,000.00	$ 66,150.00	$ 69,458.00	$ 72,931.00	
Administrative Expenses	$ 8,000.00	$ 30,000.00	$ 31,500.00	$ 33,075.00	$ 34,729.00	$ 36,465.00	
Total Costs (Future Value)	$ 346,000.00	$ 336,000.00	$ 347,550.00	$ 363,378.00	$ 380,774.00	$ 399,525.00	
Total Costs (Present Value)	**$ 346,000.00**	**$ 326,213.59**	**$ 327,599.21**	**$ 332,542.35**	**$ 338,312.77**	**$ 344,633.77**	**$ 2,015,301.69**
Benefits							
Monthly Subscriptions	$ 180,000.00	$ 198,000.00	$ 217,800.00	$ 239,580.00	$ 263,538.00	$ 289,891.80	
Annual Subscriptions	$ 120,000.00	$ 132,000.00	$ 145,200.00	$ 159,720.00	$ 175,692.00	$ 193,261.20	
Pay-Per-Lesson	$ 90,000.00	$ 99,000.00	$ 108,900.00	$ 119,790.00	$ 131,769.00	$ 144,945.90	
Non-Subscriber Sessions	$ 35,000.00	$ 38,500.00	$ 42,350.00	$ 46,585.00	$ 51,243.50	$ 56,367.85	
Subscriber Sessions	$ 30,000.00	$ 33,000.00	$ 36,300.00	$ 39,930.00	$ 43,923.00	$ 48,315.30	
Ad Revenue	$ 15,000.00	$ 16,500.00	$ 18,150.00	$ 19,965.00	$ 21,961.50	$ 24,157.65	
Enterprise Packages	$ 20,000.00	$ 22,000.00	$ 24,200.00	$ 26,620.00	$ 29,282.00	$ 32,210.20	
Affiliate Revenue	$ 10,000.00	$ 11,000.00	$ 12,100.00	$ 13,310.00	$ 14,641.00	$ 16,105.10	
Total Benefits (Future Value)	$ 500,000.00	$ 550,000.00	$ 605,000.00	$ 665,500.00	$ 732,050.00	$ 805,255.00	
Total Benefits (Present Value)	**$ 500,000.00**	**$ 533,980.58**	**$ 570,270.53**	**$ 609,026.77**	**$ 650,416.94**	**$ 694,620.04**	**$ 3,558,314.86**
Present Value Discout Rate	**3%**						

Cost Benefit Analysis	
Total PV Benefits	$ 3,558,314.86
Total PV Costs	$ 2,015,301.69
NET BENEFIT	**$1,543,013.17**

Five-year projection at a 3% discount rate. Net benefit: $1,543,013.17.

PROJECT PLAN

Gantt chart

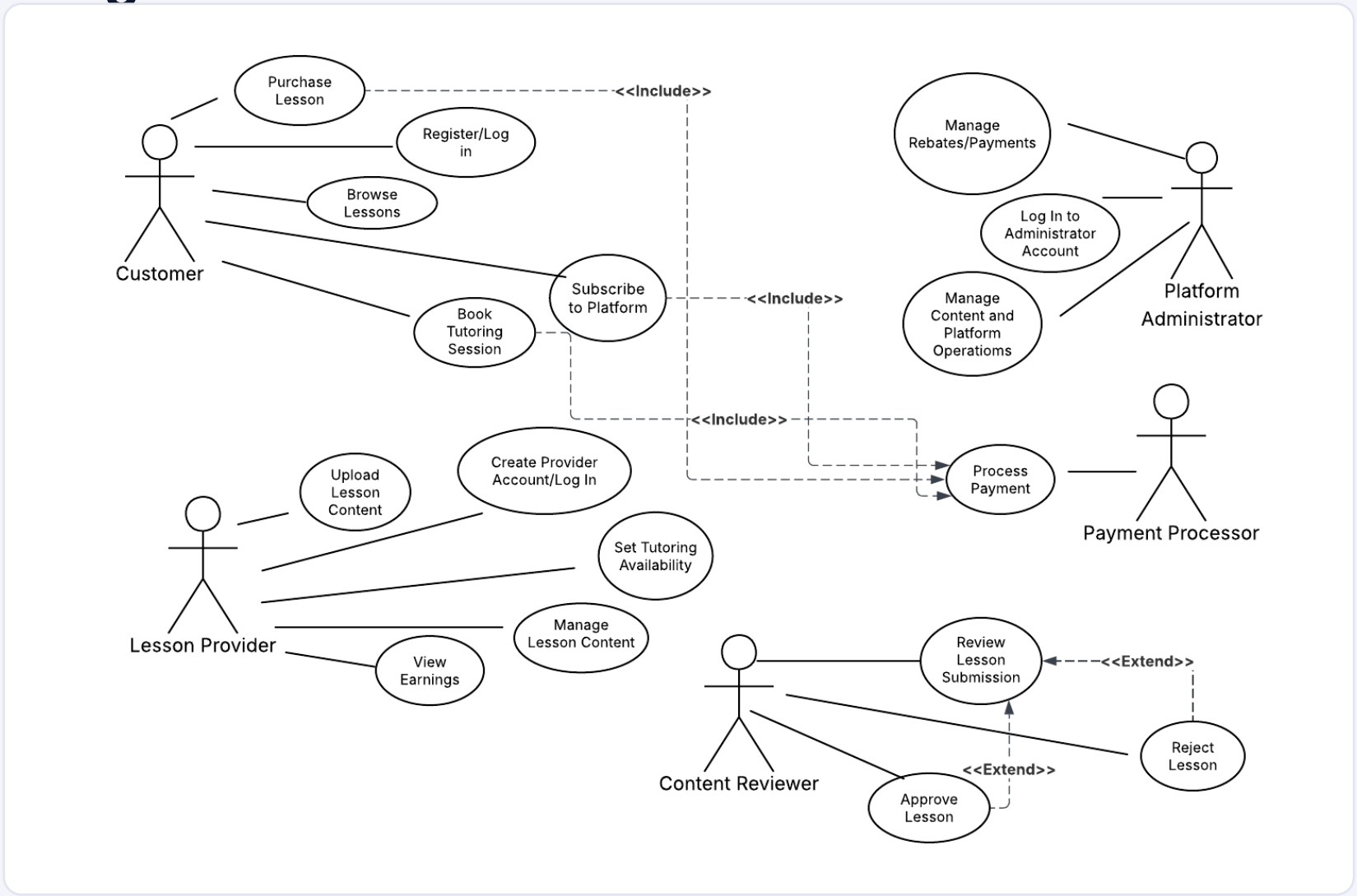
WORK BREAKDOWN STRUCTURE WITH GANTT CHART

PROJECT TITLE	TradeMind Online Learning Platform Development
PROJECT MANAGER	Supriya Shrestha
COMPANY NAME	TradeMind
DATE	1/1/26

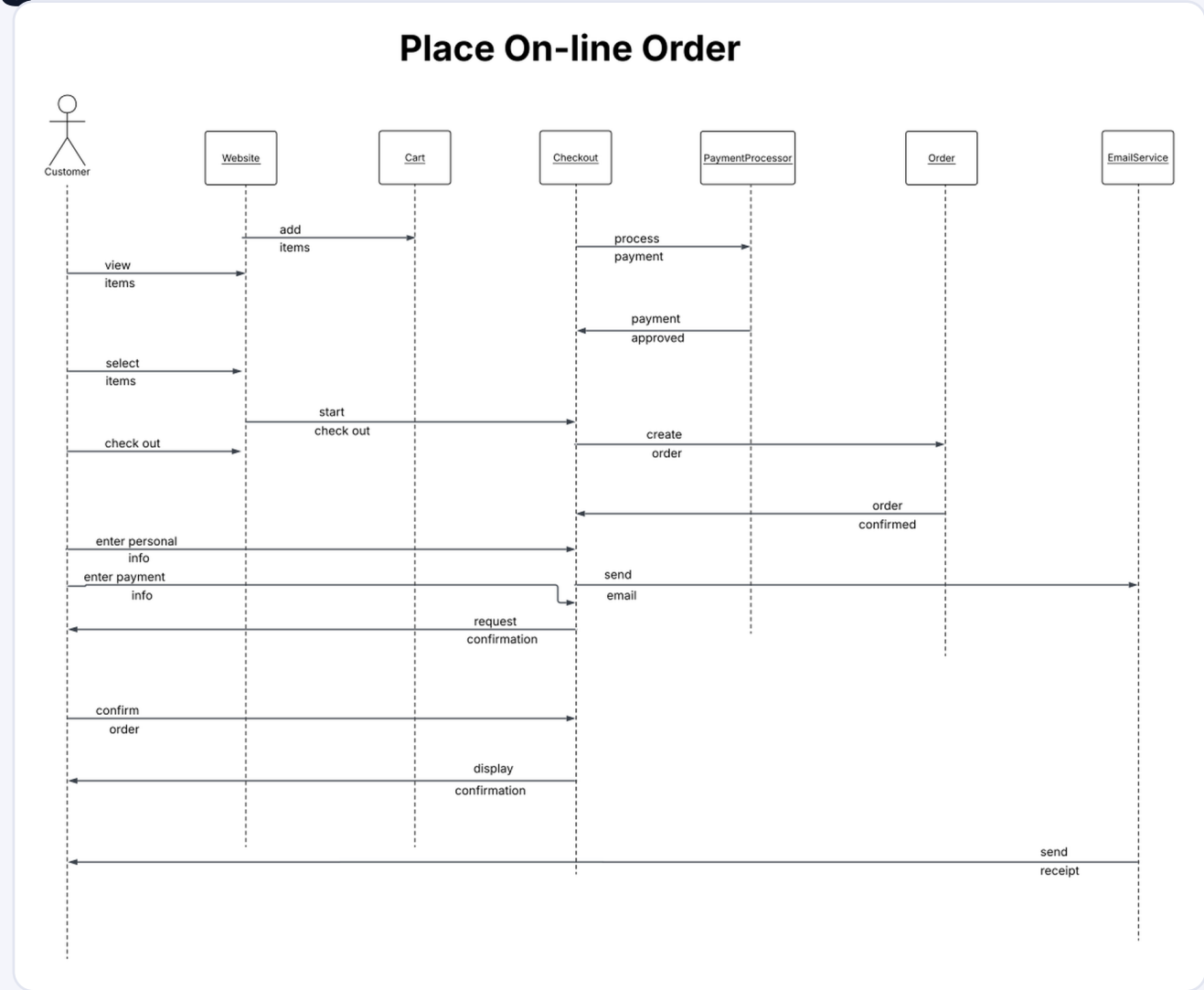


WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION	% of TASK COMPLETE
1	Inception					
1.1	Stakeholder Identification	Supriya S	1/1/26	1/5/26	4	100%
1.2	Requirements Research	Supriya S	1/6/26	1/15/26	9	100%
1.3	Conduct Interviews and Observations	Supriya S	1/16/26	1/22/26	6	100%
1.4	Feasibility Analysis	Supriya S	1/23/26	1/30/26	7	100%
1.5	Resource Identification	Supriya S	1/31/26	2/5/26	5	100%
1.6	Schedule Timeline Development	Supriya S	2/6/26	2/12/26	6	100%
1.7	Team Charter	Supriya S	2/13/26	2/18/26	5	100%
1.8	Business Case Submission/Cost-Benefit Analysis	Supriya S	2/19/26	2/26/26	7	100%
1.9	Preliminary Project Plan	Supriya S	2/27/26	3/5/26	6	100%
2	Elaboration					
2.1	Define System Scope	Vinay V	3/6/26	3/12/26	6	100%
2.2	Functional Requirements Gathering	Vinay V	3/13/26	3/20/26	7	100%
2.3	Use Case Diagram	Vinay V	3/21/26	3/27/26	6	100%
2.4	Class Diagram	Vinay V	3/28/26	4/3/26	6	100%
2.5	Sequence Diagram	Vinay V	4/4/26	4/10/26	6	100%
2.6	State Chart Diagram	Vinay V	4/11/26	4/16/26	5	100%
2.7	Data Model Diagram	Vinay V	4/17/26	4/24/26	7	100%
2.8	User Interface Design	Vinay V	4/25/26	5/1/26	6	100%
2.9	System Proposal	Vinay V	5/2/26	5/6/26	4	100%
3	Construction					
3.1	Platform Setup (AWS, Environment, repo)	Sydney R	5/11/26	5/18/26	7	10%
3.2	User Accounts and Authentication	Sydney R	5/19/26	5/30/26	11	0%
3.3	Lesson Upload and Content Management	Sydney R	5/31/26	6/12/26	12	0%
3.4	Payments and Subscriptions	Sydney R	6/13/26	6/22/26	9	0%
3.5	Approval Workflow	Sydney R	6/23/26	6/29/26	6	0%
3.6	Tutoring Sessions (Booking + Scheduling)	Sydney R	6/30/26	7/10/26	10	0%
3.7	Testing and Bug Fixes	Sydney R	7/11/26	7/22/26	11	0%
4	Transition					
4.1	Production Deployment & System Go-Live	Cherish A.	7/23/26	7/30/26	10	0%
4.2	Data Migration & System Validation	Cherish A.	7/31/26	8/7/26	7	0%
4.3	Staff Training (Operations, Content Reviewers, Customer Support)	Cherish A.	8/8/26	8/15/26	8	0%
4.4	Marketing Launch & Creator Onboarding	Cherish A.	8/16/26	8/31/26	15	0%
4.5	Post-Implementation Support & System Evaluation	Cherish A.	9/1/26	9/20/26	20	0%

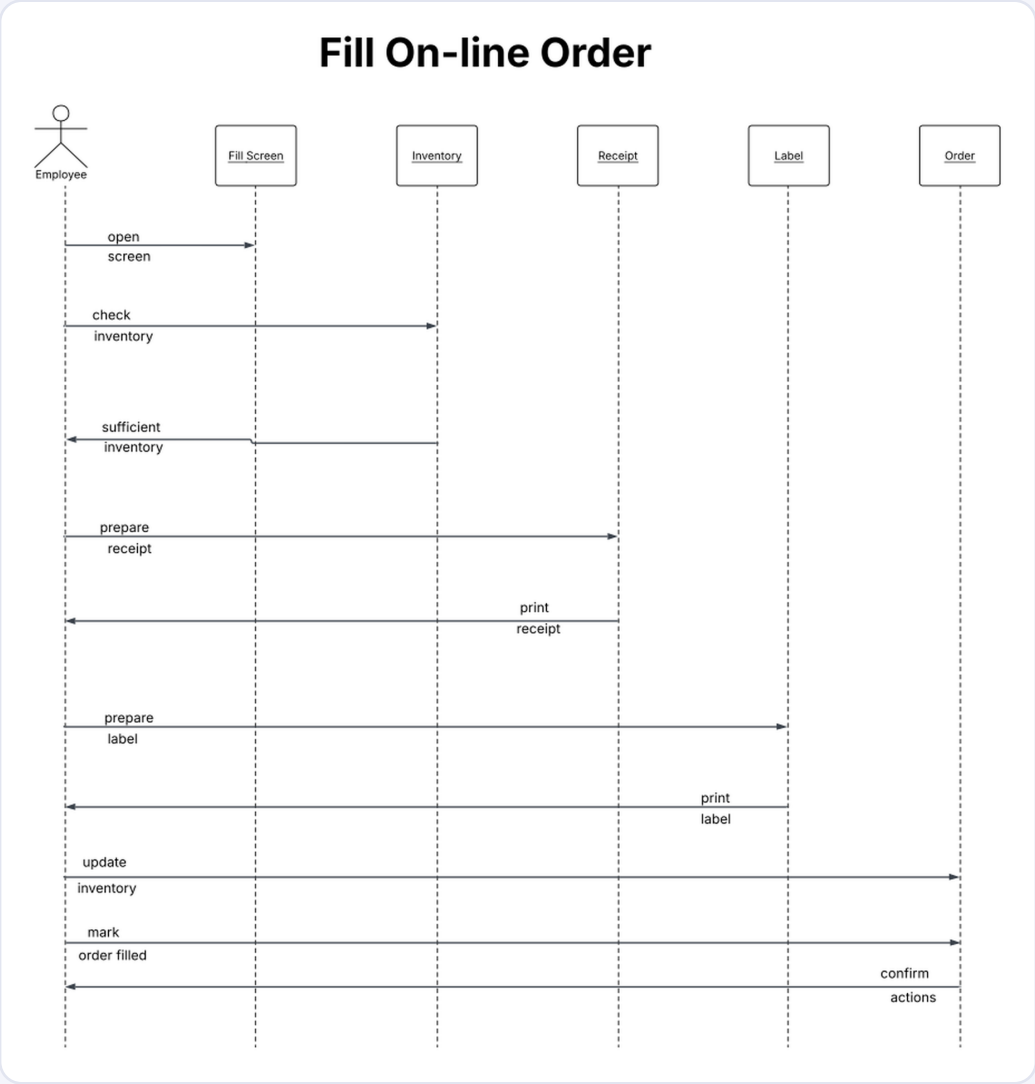
Use case diagram



Sequence diagram: Place on-line order

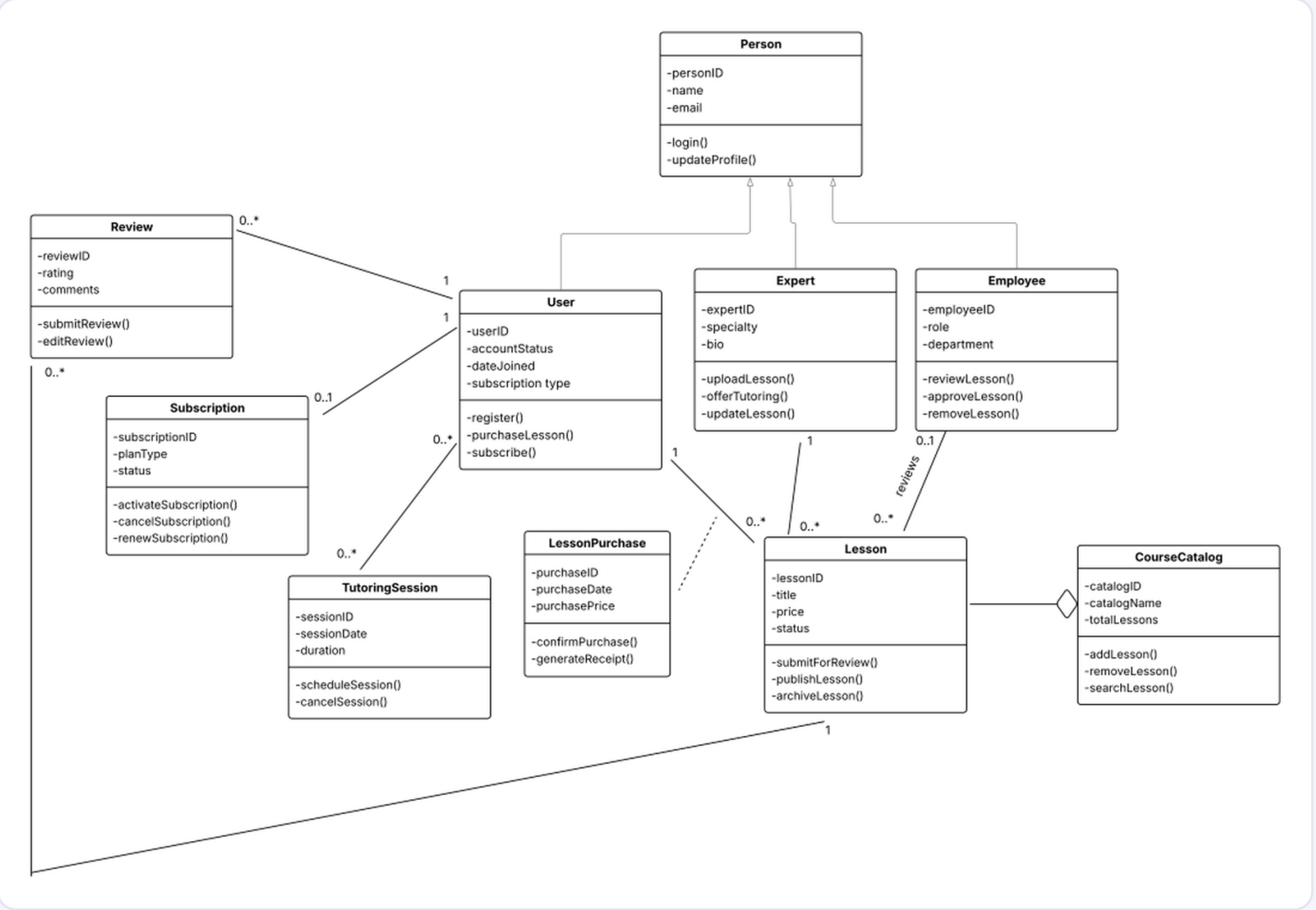


Sequence diagram: Fill on-line order

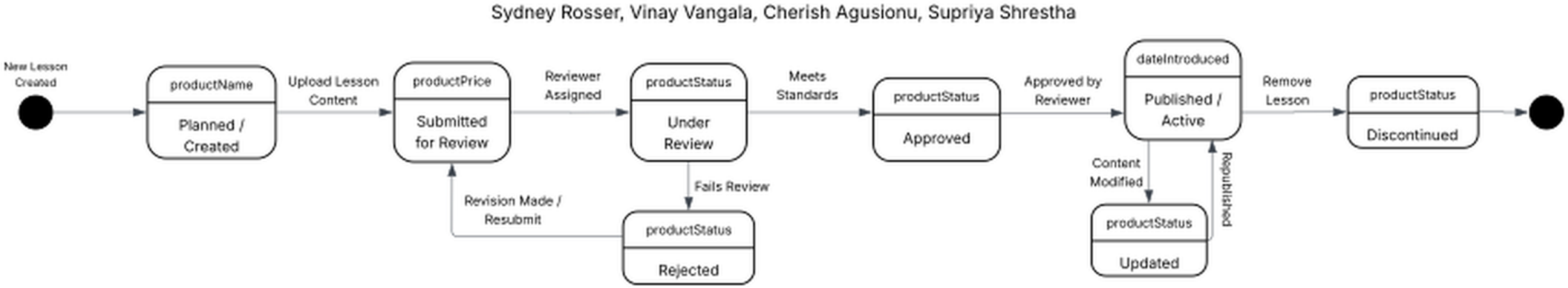


UML MODELS

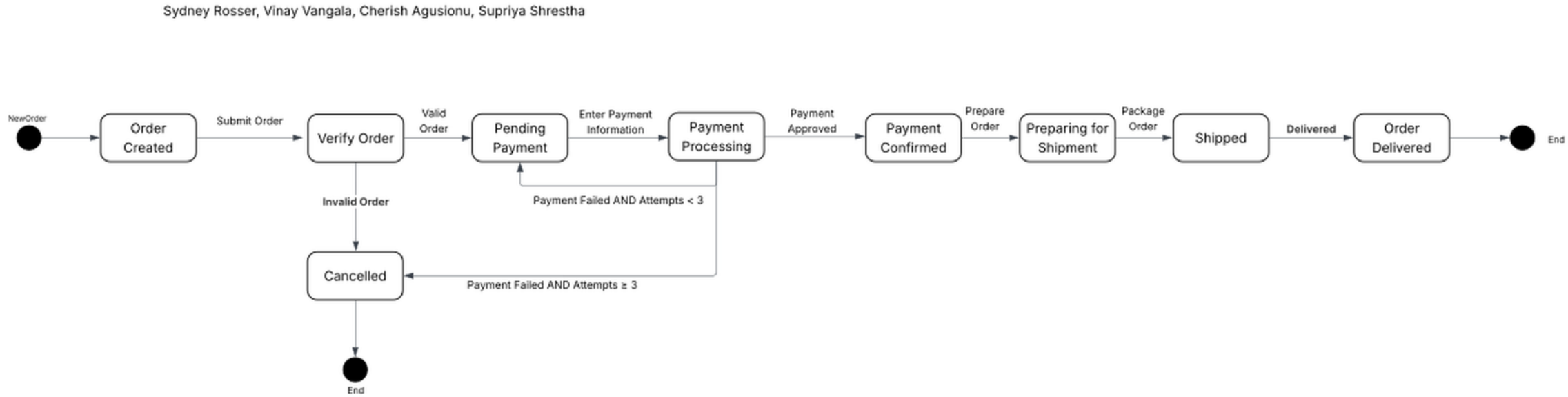
Class diagram



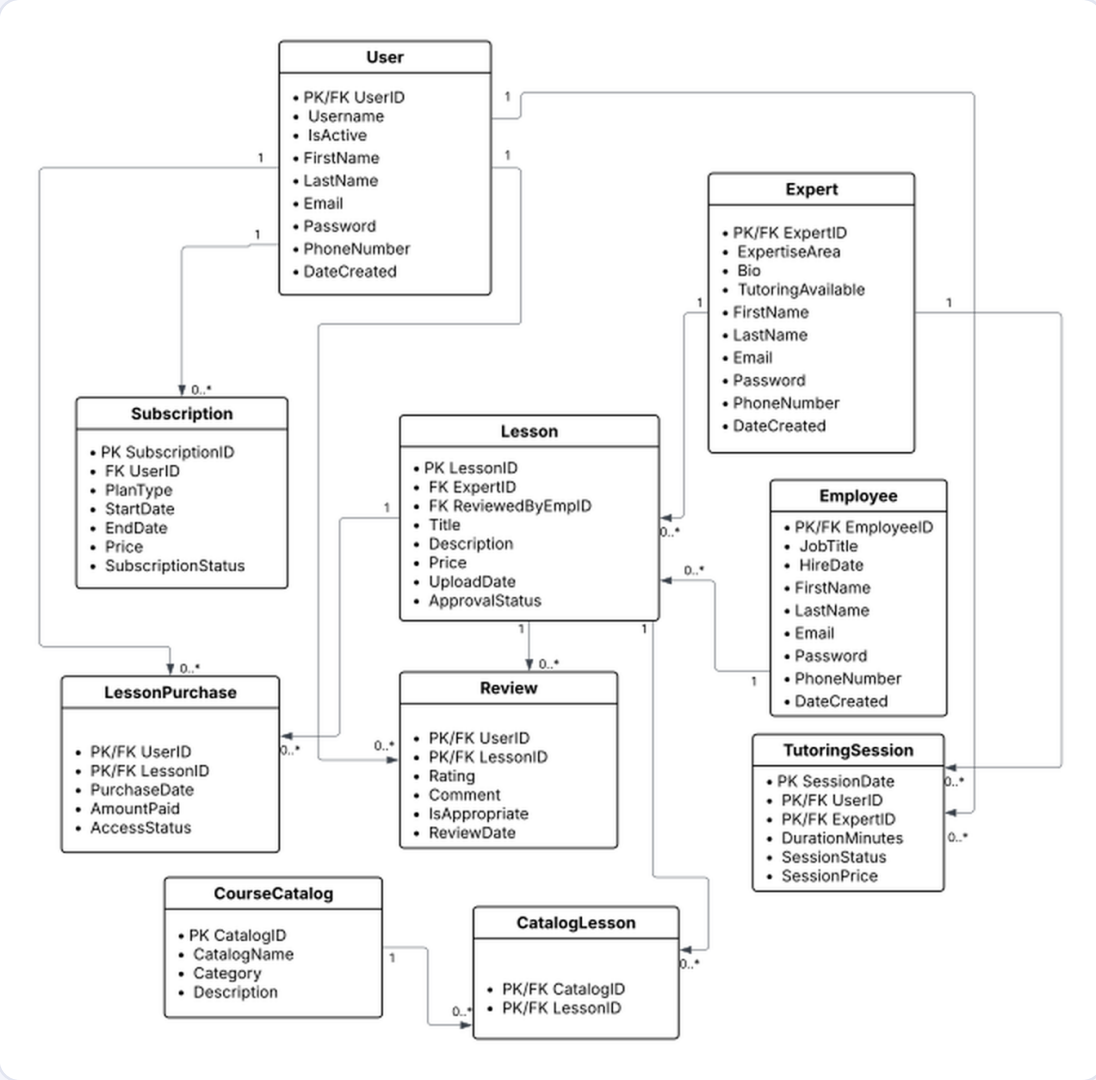
State chart: customer order



State chart: product

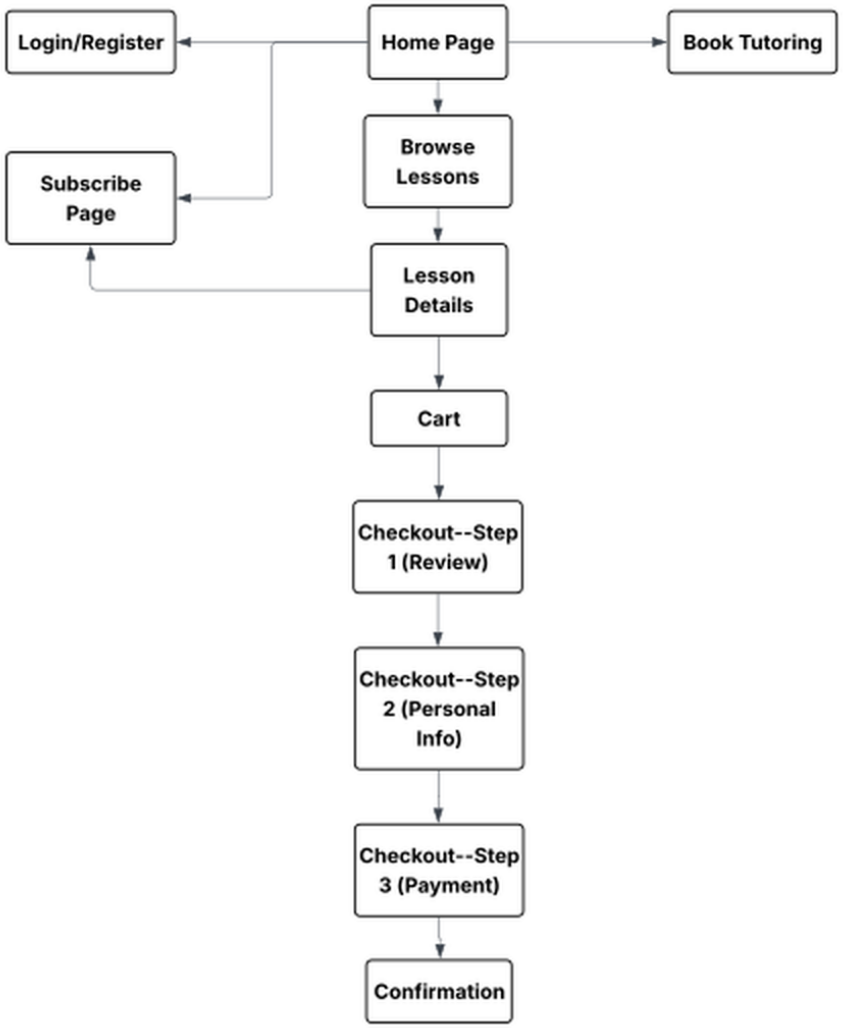


Normalized data model



DATA AND NAVIGATION

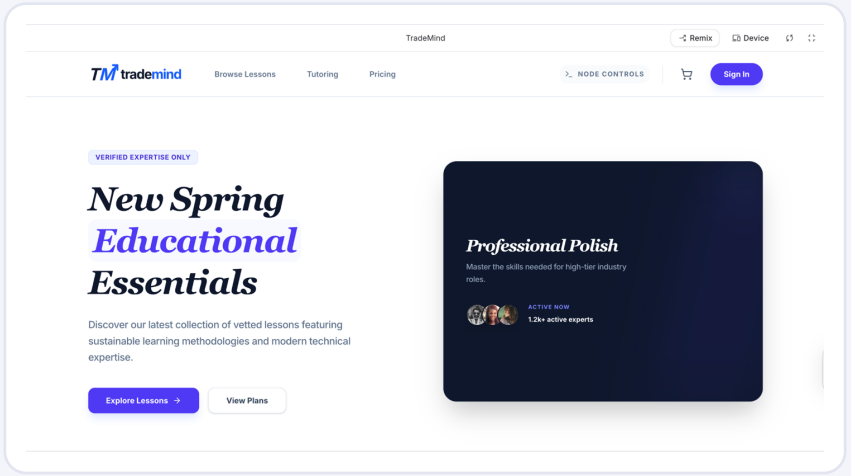
Windows navigation diagram



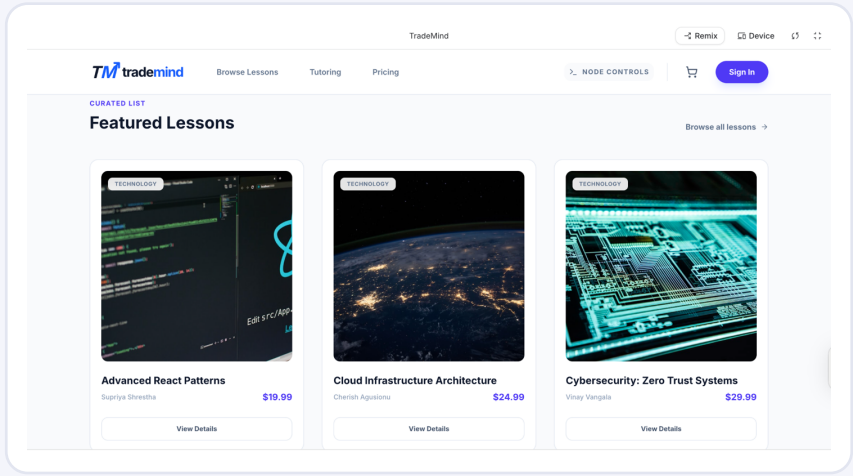
How users move through the site

From login or registration, users reach the home page, then branch to browsing lessons, subscribing, or booking tutoring. Lesson details lead to the cart and a three-step checkout (review, personal information, payment), ending at confirmation.

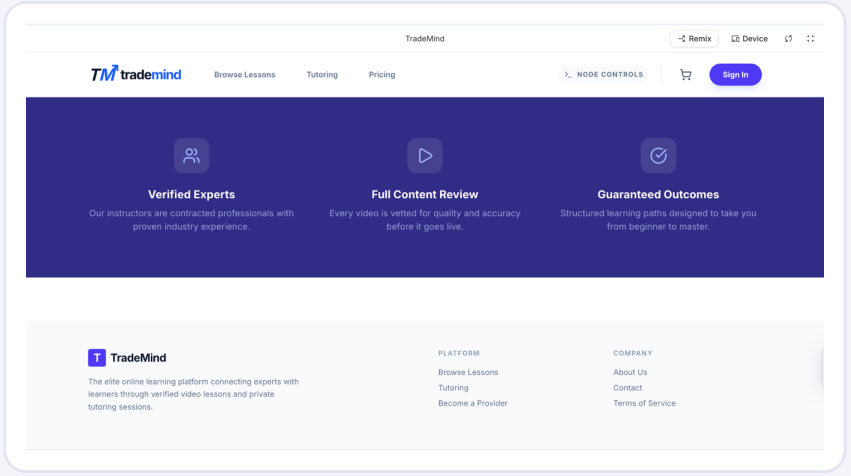
Interface prototypes (1 of 4)



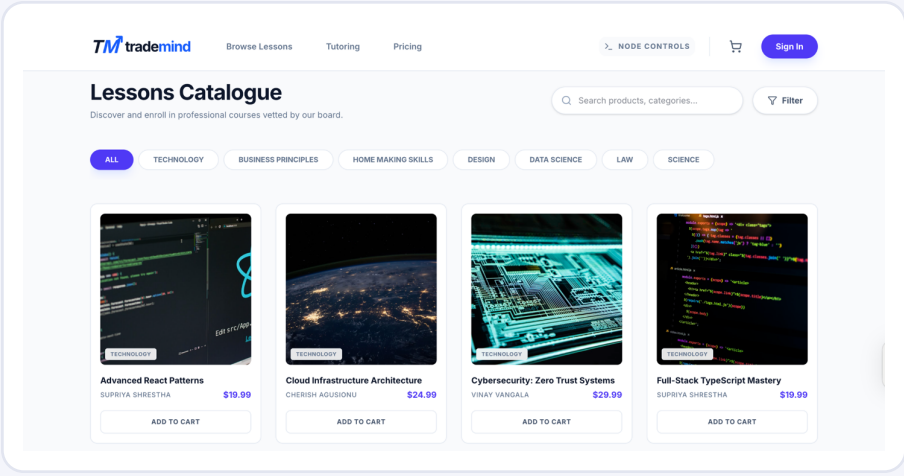
Home page



Featured lessons

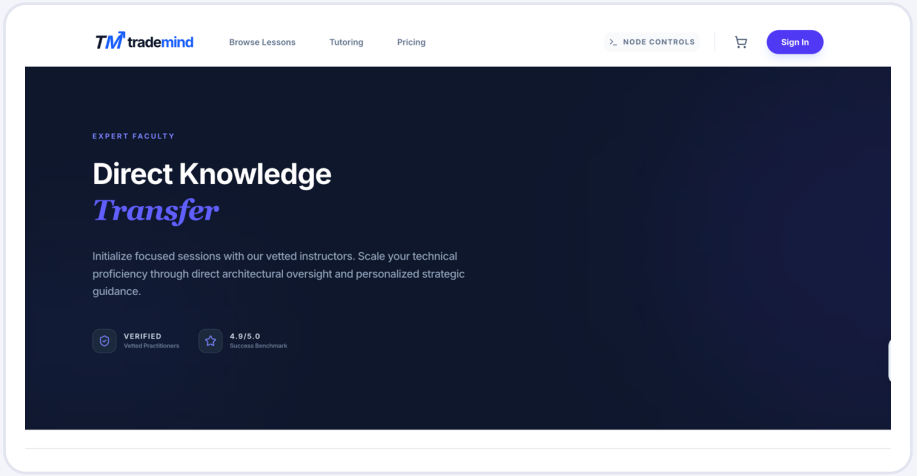


Value propositions and footer

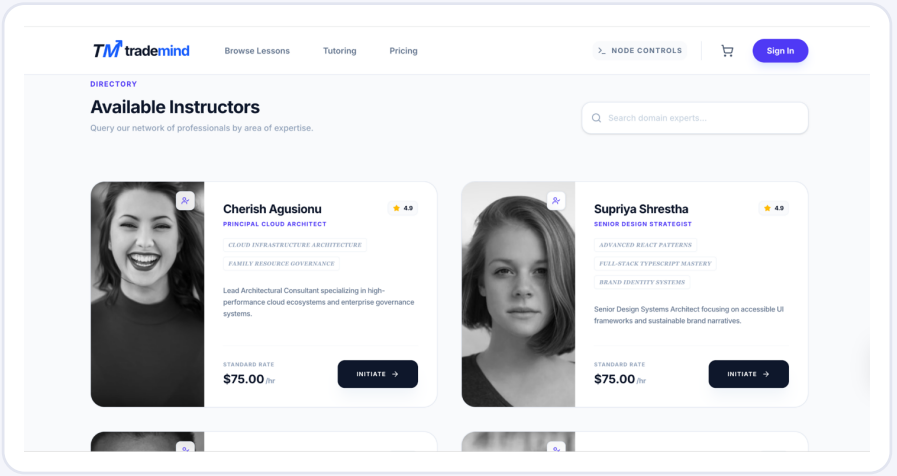


Lesson catalogue with search

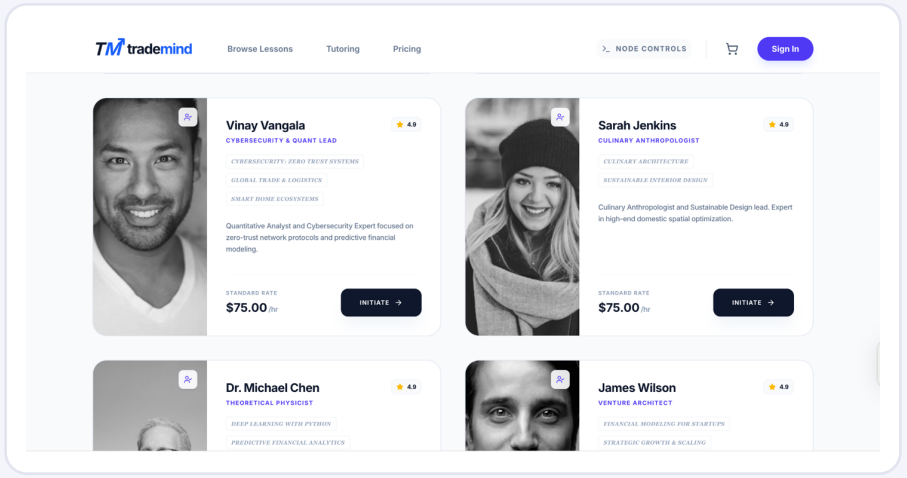
Interface prototypes (2 of 4)



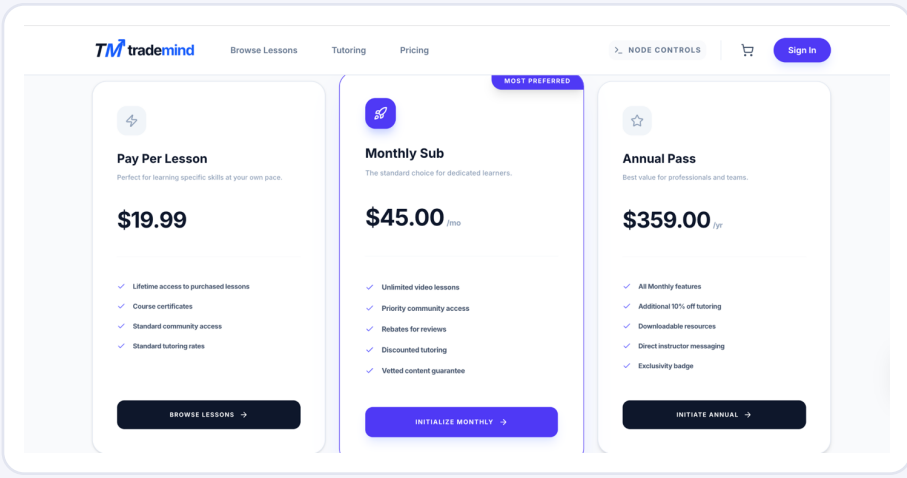
Expert faculty



Instructor directory

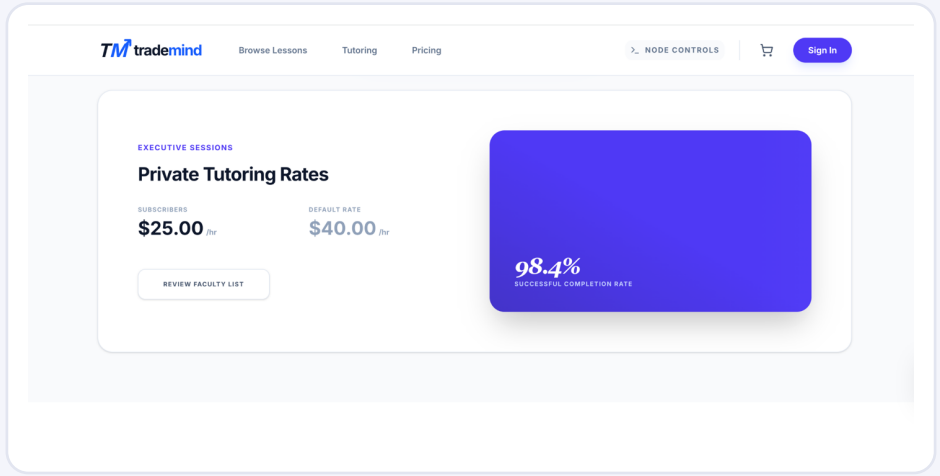


Instructor profiles

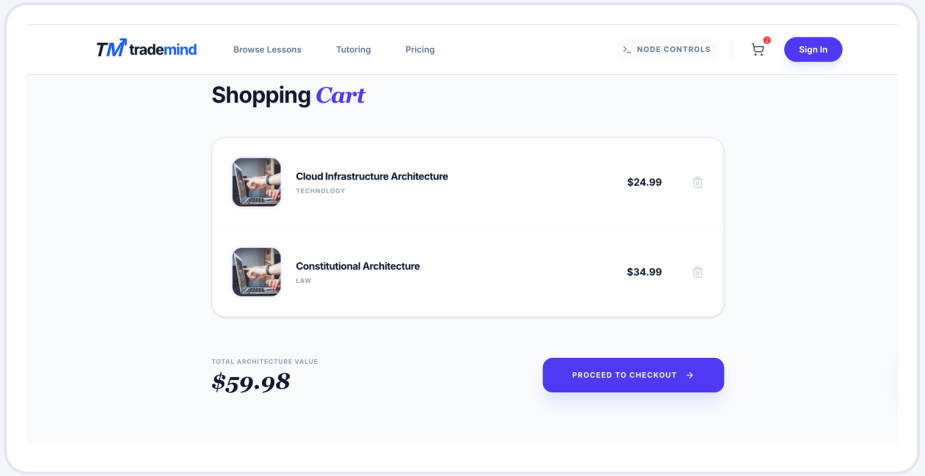


Pricing plans

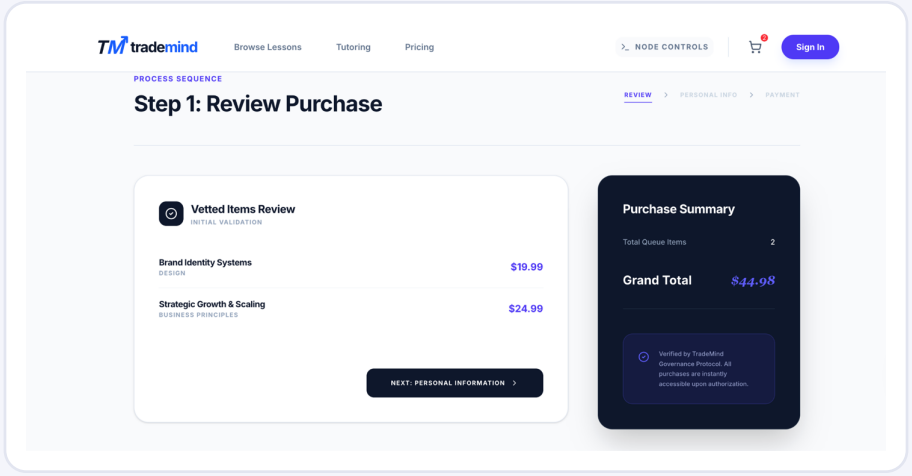
Interface prototypes (3 of 4)



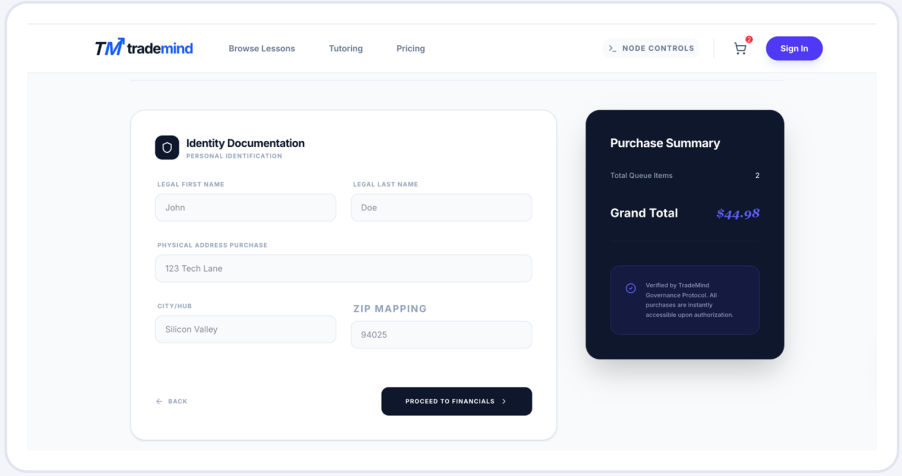
Private tutoring rates



Shopping cart

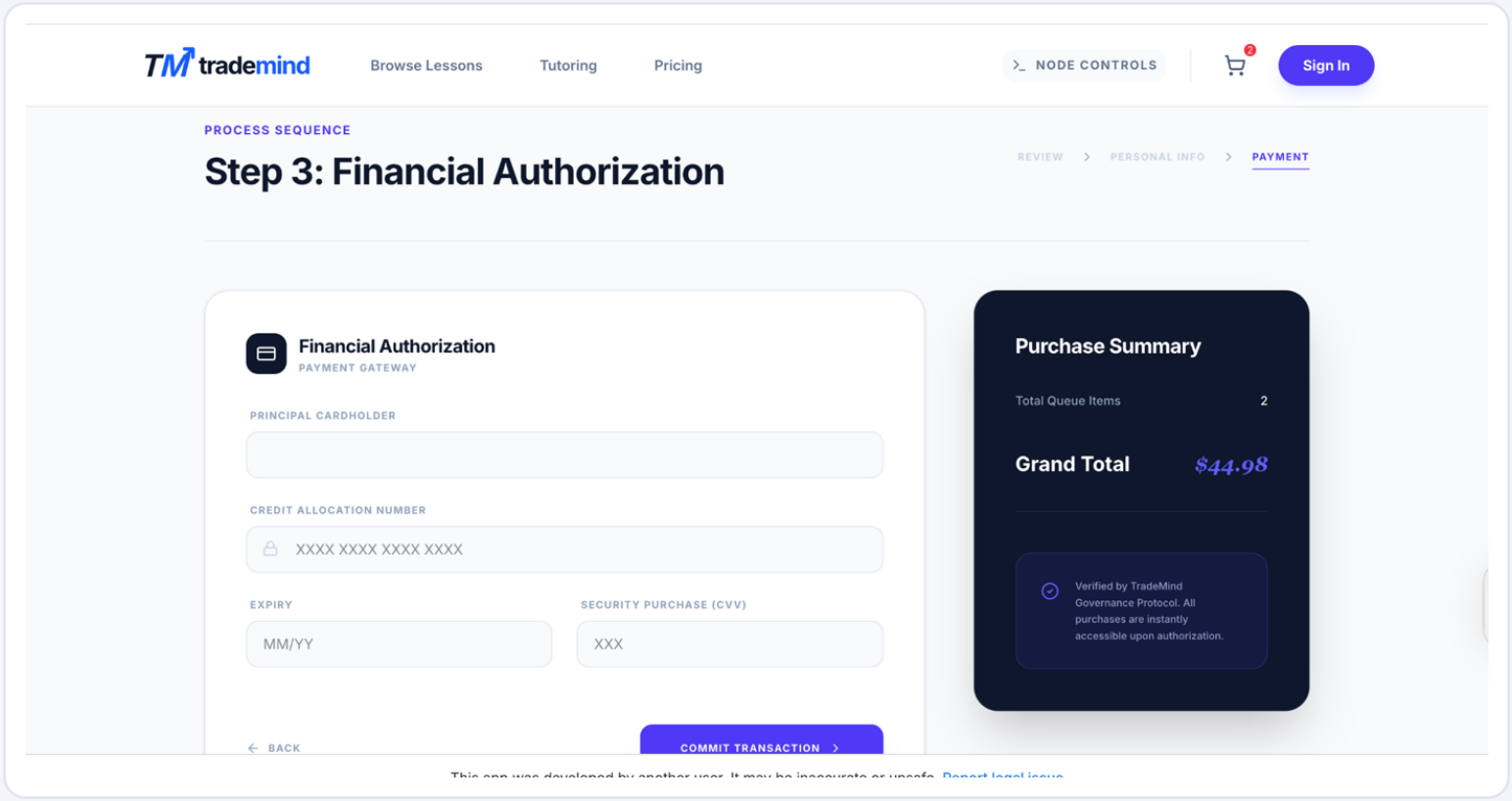


Checkout step 1: review purchase



Checkout step 2: personal information

Interface prototypes (4 of 4)



Checkout step 3: financial authorization

The final step of the multi-step checkout, with a purchase summary beside the payment form.

USABILITY REPORT

What testers said

Two people outside the design process, Ariel Citizen and Fey Ojo, reviewed the screens and gave feedback on first impressions and navigation. Tasks: identify the site’s purpose, browse the catalog, review instructors, compare pricing, and move through the cart and checkout.

What worked

- Both immediately understood TradeMind as a premium learning platform
- Described as clean, professional, and visually appealing
- Consistent branding and organized content read as trustworthy
- Lesson cards, instructor profiles, and pricing were easy to understand

What they suggested

- Make homepage call-to-action buttons more prominent (Ariel)
- Make catalog search and filters stand out more (Ariel)
- Add clearer progress indicators between checkout steps (Fey)

What changed

- Greater visual emphasis on key action buttons
- Review of search and filter prominence
- Stronger step-progression cues through checkout
- Clean aesthetic preserved

EXECUTIVE SUMMARY

Ready for Construction

Where the project stands

- **Inception complete:** business case, feasibility analysis, cost-benefit analysis, project schedule, requirements research
- **Elaboration complete:** use case, class, sequence, and state chart diagrams; normalized data model; windows navigation; interface prototypes; usability report
- **Next, Construction:** developers build the web platform, lesson streaming, scheduling, moderation, security, and database tools
- **Then, Transition:** deployment, administrator training, expert onboarding, monitoring, and support

Recommended after launch requirements

- Optimize search and lesson recommendation algorithms
- Develop a mobile TradeMind app
- Build more robust analytics dashboards
- Improve accessibility and translation features
- Review and update security measures continually

Requirements are well defined and the project is on schedule.